



Pokhara University
Everest Engineering College
Final Internal Assessment
Fall-2025

Level: Bachelor

F.M. 100

Program: BE CMP(7th Sem)

P.M. 45

Section: A/B

Time: 3hrs

Subject: Data Science and Analytics

Attempt all the questions.

- 1 a) Explain different data cleaning process. 7
 b) Explain Knowledge Discovery in databases. A numeric variable has the following values. 21.5, 67.8, 42.7, 8.3, 102.55, 57.89, 22.77. Standardize the values. 8
- 2 a) Explain and interpret Zipf's law. Also list application and limitation of Zipf's law. 7

Age (x)	Salary (y, in \$1,000s)
25	30
32	45
40	55
48	65
55	75

8

Calculate and interpret Pearson's co-relation coefficient. Explain the purpose of T-value, P value and ANOVA test.

- 3 a) For the given dataset, find the regression equation using matrix approach. 8

Observation	x_1	x_2	y
1	1	2	6
2	2	1	8
3	3	3	12

7

- b) Explain R^2 and adjusted R^2 in the context of model evaluation in detail.

OR

Derive the equation of Nadaraya-Watson kernel regression estimator.

- 4 a) Given data = {2,3,4,5,6,7;1,5,3,6,7,8}. Compute the principal component using PCA Algorithm. 8

- b) Given a new dataset to demonstrate non-parametric regression: 7

$$X = [1,2,3,4,5]$$

$$Y = [2,4,1,5,3]$$

Estimate $\hat{y}$ at $x = 2.5$ using the Nadaraya-Watson kernel regression with Gaussian kernel:

$$\hat{y}(x) = \frac{\sum_{i=1}^n K_h(x - X_i) Y_i}{\sum_{i=1}^n K_h(x - X_i)}$$

where $K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$ and bandwidth $h = 1$.

- 5 a) Describe CART algorithm for decision trees. How are splits determined in classification? 7

- b) Make a decision tree from the following dataset. Assume necessary data and algorithm. 8

Outlook	Temperature	Humidity	Windy	PlayGolf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

2 -

7/6

5/2/5

OR

Cluster the following eight 2D points using the K Means algorithm with $K=3$ and perform 2 iterations.

Point	Coordinates
A_1	(2,10)
A_2	(2,5)
A_3	(8,4)
A_4	(5,8)
A_5	(7,5)
A_6	(6,4)
A_7	(1,2)
A_8	(4,9)

- 6 a) Explain the components of time series analysis in details. 7
b) Briefly describe ARMA and ARIMA models. How do you select the optimal lag length in ARIMA model? 8
- 7 Write short notes on: (Any two) 2*5=10
a) Cross tabulation
b) Logistic Regression
c) Causation Vs Correlation

****Good Luck****